An Approach to Student Experience and Knowledge Development in Rocket Propulsion and Support Systems

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***Abstract* – The Experimental Rocket Propulsion Lab (ERPL) at Embry-Riddle Aeronautical University focuses on student-run, undergraduate research that offers members the chance to develop key skills and knowledge by designing, building, and testing rocket subsystems. Members are encouraged to submit ideas for new projects so that they can develop experimental propulsion and support systems. Current projects include a modular hybrid research engine, an integrated flight hybrid engine, a liquid rocket engine, computational fluid dynamics research, a guided rocket parachute, as well as several others. Each of the projects offers a detailed insight to a specific component of a rocket system. Practicing developmental understanding through experiments has become a crucial way ERPL builds its members' knowledge on subsystems operation in the world of rocketry.**

Keywords – Undergraduate research, model-based systems, simulations, & systems verification.

I. Introduction

ERPL is a student organization involved in the research, development, and production of rocket engines and support systems. Comprised of many projects, the club is made up of several divisions that work simultaneously in the same lab by students with a broad range of majors and skill sets. Currently, active divisions include hybrids, liquids, landing systems, and simulations. While all of these divisions are technically separate, they often work in conjunction and are used to support one another. Furthermore, all divisions are open to members of any skill level. Members are also encouraged to submit their own project ideas in order to be supported by ERPL.

In addition to producing advancements in rocketry and increasing industry understanding, ERPL is centered on providing student members with critical project experience. Each semester more than 50 active members participate in a growing club that spends each year offering more projects for members to be exposed to, making ERPL a diverse organization. New members are offered the chance to participate in a high powered rocket certification program created by ERPL in 2014. Many members build their own rocket for the time and receive their high power certification via the National Association of Rocketry. This program develops members' fundamental rocket skills so that they can further grow and assist in project development.

Discussed in this paper are a few key projects undergoing current development. Each project focuses on a highly detailed subsystem based around rocketry and propulsion. Not only are these projects complex and difficult, but they offer many of the real-world challenges that are faced every day in various aerospace industries.

In many studies on undergraduate experience in high-level research projects, students flourished because they were able to combine conceptual education they get in class supplemented with experimental education from research projects [1, 2, 3]. Upperclassmen typically fill leadership positions, directing projects while underclassmen and less experienced members are guided while developing a project. The exchange elicits the development of the leaders' communication and managerial skills and the apprentice to build upon base knowledge in a real-life experience. In the end, all of this experience is strongly applicable to building a career in the aerospace industry. [4].

However, the lack of experience and background knowledge on subject matter poses a problem for undergraduate students seeking to solve complex research problems. Therefore, ERPL uses experienced researchers and other knowledgeable contacts to fill the gap. These professional contacts will serve as life-long contacts for leaders and heavily-involved student members when working on future projects. This fosters the opportunity for life-long learning [5].

II. Hybrids Division

ERPL is currently in the testing phase for two hybrid engines. Both engines use liquid nitrous oxide as the oxidizer, and hydroxyl-terminated polybutadiene (HTPB) as the fuel.

A. Projects Hydrus & Kryso

ERPL's first hybrid engine, Hydrus, began development in 2009. Hydrus is a compact hybrid engine that features an integrated oxidizer tank and injector with external ignition. The 29 millimeter engine provides a near-instantaneous thrust of 80 Newtons. The majority of the volume of the engine is comprised of the integrated oxidizer tank. Practicality limits the engine to single port fuel grains because of its small diameter. The all-aluminum engine will be the powerhouse for the carbon-fiber/fiberglass rockets

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being created. These rockets are designed for low altitude flights with missions ranging from engine test beds to dynamic flight control integrated into the rocket construction. The students behind the design of the engine have long since graduated; however, the new team has pushed the engine through the manufacturing phase, and is currently prepping the ground support equipment for the testing phase. Current objectives including refitting an old test stand with new electronics, verifying the structural integrity of a new combustion chamber and redesigning the umbilical support system.

While working on this project, younger students have been exposed to a great deal of knowledge from the previous generation of students and advisors. During the development of Hydrus, members gained insight into manufacturing processes that are not typically taught inside the classroom. When manufacturing the engine, students were able to see what machines would be making the parts they designed. This allows students the ability to learn the specialties and limitations of typical machine shop equipment, and incorporate it into new designs. From this knowledge, parts have since been revised to make the machining process a simpler and smoother one in order to get the parts moved quickly from drawing to test stand. Geometric dimensioning and tolerancing (GD&T) practices and machine knowledge were taught throughout the manufacturing process. The design classes many students take emphasize theory and precise values, however that does not translate well to actual machine shop practices. Knowledge of the tools being used to create engine components greatly affects how the drawing is submitted for manufacture. This hands-on project gave students insight as to what compromises can be made in the design without sacrificing performance, with the benefit of decreasing time spent in the machine shop.

Figure 1: The composite roll-up process for producing the body tube of Kryso

While the engine is being developed, a new rocket has been designed to fly this engine based on the designs of previous rockets flown by the Hybrids Division. A new rocket called Kryso, made from carbon fiber and fiberglass, is shown being constructed by students in Figure 1. Thanks to a malfunction that occurred in the parachute deployment system during experimental testing, the composite body tube design has been proven to survive a mile high fall.

Kryso is a vital component of the Hybrid's Division, as it is currently the only flight vehicle in development. In addition to being important for practical mission success, it is also a springboard for future leaders of the division. As the fuselage is more forgiving than the engine, the responsibility of the fuselage and ground support systems is traditionally given to young members. ERPL is built on a self-starting idealism, and Kryso has been an opportunity for students to get experience with rockets as well as with leadership. Out of the team of four freshmen and sophomores that manufactured and tested Kryso, one of them has moved on to be the project lead for Hydrus while the other is the Hybrids Division lead. The current team for the Kryso ground support system is comprised of one freshman and one sophomore. Due to the leadership and ability to take charge of a task being shown from working on Kryso, one of them will be moving up to Hydrus project lead in the very near future.

B. Project Vulcan

The second hybrid engine carries the name Vulcan, and it is an experimental ground test engine. Vulcan features a modular engine design, meaning every component of a rocket engine has been machined individually with interlocking features. This modular design allows Vulcan to be modified or tailored for specific applications without re-designing and re-machining the entire engine. The modular design of Vulcan means that the engine can alter its thrust output from 220 Newtons to 2000 Newtons simply by swapping one nozzle component with another. In order to vary burn times and thrust profiles, multiple fuel grain configurations can be created in the 130 mm internal diameter engine. The current configuration is optimized for hydroxyl-terminated polybutadiene (HTPB) and nitrous oxide. For reference, a single port fuel grain of HTPB has an expected burn time of 10 seconds. However, the versatility afforded by Vulcan allows for many different fuel configurations to be tested in the same engine. Since the same fuels being used for both Vulcan and Hydrus, the same external ignition system can be used for both engines. Like Hydrus, Vulcan is also in the testing phase of development. Two successful hot fires of Vulcan have been conducted thus far, and the results of which have prompted alterations to the design to create an even more versatile engine.

While working on Vulcan, members are able to gain experience with a functional rocket engine and its relevant systems. For many students currently in ERPL, Vulcan is the first partially liquid rocket engine they have seen perform a static fire during their entire time at the University. This gives the younger members experiences which they can use to frame equations and theories that they normally only see in the classroom. For example, it is much easier for the members to understand the importance of a hoop stress calculation on a combustion chamber upon receiving firsthand experience for themselves. Much of the knowledge students gain at the University is knowledge of equations and theories, and ERPL bridges the gap from the classroom to industry by offering students a chance to apply the theory and exercise their brains outside of the classroom with real equipment and technology.

III. Liquids Division

A. Project Aquila

Aquila is the overarching name given to two separate liquid rocket engines: EV-1 and EV-2. The purpose of both

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of these engines is to give interested members of ERPL the opportunity to get hands on with two-part liquid engine design and operation. The team members will be expected to utilize proper documentation and procedures to ensure that safe, reliable engines are produced. Team members will also develop skills in safety analysis, CAD design, mechanics, propellant storage and utilization, electronics, fluid analysis, operation of liquid engines and teamwork.

EscapeVelocity-1 (EV-1) is a small system that has required an overhaul and some modification since its original design to convert it from a two-part liquid system to a gaseous oxygen (GOX), liquid fuel (propane-C3H8) system. The use of GOX and a petroleum-based fuel create an easy and relatively inexpensive propellant for regular firing. An already complete fuel pressurization system utilizes nitrogen as the pressurant to push the liquid fuel. The fuel pressurization system consists of simple components like the nitrogen tank, manually operated regulator and blow off valves; and electrical components like solenoid valves and pressure transducers for long distance operation and monitoring. Figure 2 shows the EV-1 pressurization system prior to the GOX modifications, which consist of removing the oxidizer press line (blue) and capping it just after the solenoid valve.

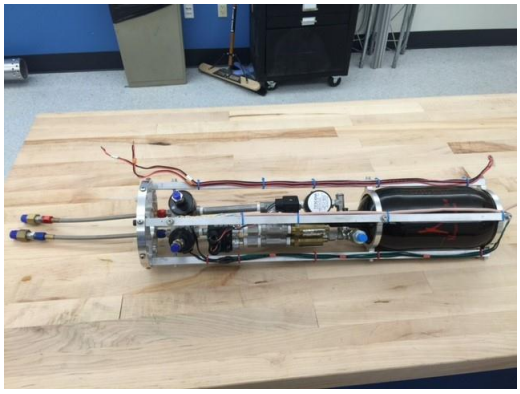

Figure 2: The EV-1 Nitrogen fuel pressurization system

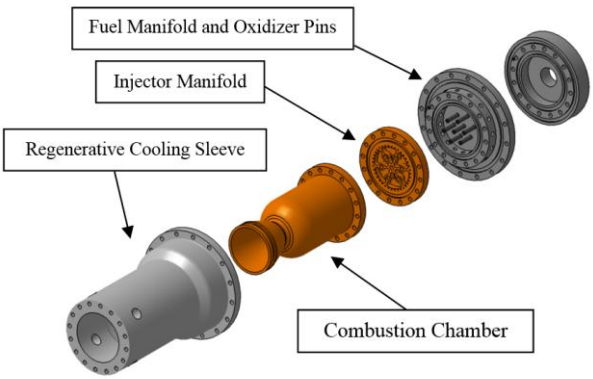

Figure 3: An exploded view of EV-1's engine components

Because the oxidizer/fuel combination has been changed as mentioned before, parameters for the volumetric and mass flow rates for both the oxidizer and fuel have been re-calculated to match the parts that have already been manufactured; namely the chamber liner. The chamber liner is made of copper for its high heat conductivity and good creep resistance. Those characteristics, combined with a regenerative cooling system, will naturally allow for extended burns, and hence, more data gained per burn. Figure 3 shows the exploded assembly of EV-1. The flow paths of the fuel and oxidizer in the complete EV-1 engine can be seen in Figure 4. The purpose of this engine has been to provide a reliable platform on which incoming students can get acclimated to liquid rocket engine design and testing.

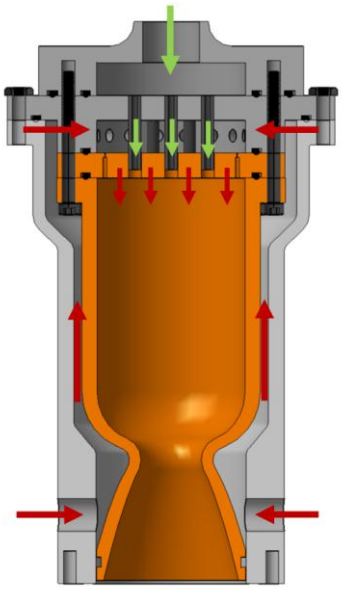

Figure 4: Computer-rendered cross section of the EV-1 Engine. Green arrows show the oxidizer flow. Red arrows show the fuel flow.

Figure 5: The machined combustion chamber and nozzle for EV-1

Due to the complexity of developing a liquid rocket engine, every member plays a crucial role in this project's development. The learning curve is very steep for all students involved, and almost all the leadership roles are

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taken by sophomores with initially limited rocketry experience. These students continue to develop their knowledge by spending time reading various design reports and technical documents about liquid rocket engines and studying independently with professors. Members have worked together to learn about fluid and thermal dynamics, stress analysis, combustion chemistry and materials compatibility. They have also worked together to develop system schematics, firing procedures and standard operating procedures for when EV-1 is tested. Parts for EV-1 are currently being acquired and cold fire tests, followed by a hot fire test, are likely to occur in the near future. Once EV-1 has been fully developed and function tested, EV-2 will become the primary focus.

EV-2 will likely be Aquila's flagship engine. It will be an almost completely "from scratch" design and build that will give the team members a complete understanding of proposal development, system design, analysis, manufacturing, assembly and testing. This engine will produce approximately 1500lb thrust and is intended to be flight capable, which adds many difficulties to the project (e.g. mass and recovery systems). EV-2 is currently expected to utilize cryogenic propellants in the form of liquid oxygen (LO2) and liquid methane (CH4). It will need to be capable of sustaining long burn durations in order to reach desired altitudes. EV-2 is the most ambitious project of Aquila, but it will provide the best research opportunities and experience for the involved team members.

IV. Landing Systems Division

A. *Project Polaris*

Recently, ERPL has broadened its scope to cover more than just experimental propulsion systems. Polaris is a project that was initiated in spring of 2015. The primary goal of this project is to develop an autonomous payload that will deploy and operate a parafoil that guides a launched rocket to a pre-determined landing coordinate. Typically, large-scale rockets use standard, circular parachutes that are liable to drift miles away from a launch site. Instead, this project will help ensure consistent, reliable recovery of a rocket every time it is launched. Furthermore, this payload will be capable of saving and delivering flight data in real time to a portable ground control station. Polaris will require the combination of several engineering disciplines. Skills including software development, unmanned autonomous systems engineering, aerodynamics, electrical engineering, mechanics, and experimental testing and analysis will all be required in order to complete this project.

The parachute system payload will consist of multiple sensors that will measure heading, altitude, GPS location, speed, and acceleration. A Pixhawk system is being used as the flight computer to collect and interpret the sensor readings. It will operate the servos connected to the parasail guiding the rocket while relaying this data through a line-of-sight data link. This data will be received by the ground control station, where position is displayed three-dimensionally on Google Maps along with heading, speed, and other important information. The ground control station will be capable of issuing manual commands to the parachute in the event of a system failure or emergency. Furthermore, there will be several safety features that will be capable of deploying a reserve parachute under specified conditions.

Figure 6: The RC Paraglider platform used for initial tests of the parachute guidance systems

In order to ensure project success, a rigorous set of testing steps have been created for this project. The initial phase of this begins with a remote control (RC) paraglider as seen in Figure 6. This paraglider is a commercial hobby aircraft that was cheap and affordable. It is being used for initial testing of the parachute's weight loading and directional response capabilities. Upon the development of an autopilot, this paraglider will also serve as the initial test platform for the autopilot. A test payload will be fitted to the paraglider, the paraglider will be flown high into the air, and the test payload will be turned on to guide the paraglider to a landing coordinate as if it were a rocket. This testing method will limit the need to launch a rocket every time the autopilot is to be tested. Furthermore, this will be safer in the event of a failure because the paraglider will not be tested nearly as high as a rocket would and would be less prone to other failures as well.

Following multiple successful tests of the paraglider and its payload, the next stage of testing will be conducted on a small-scale, 4-inch high-powered rocket. First, parachute deployment methods will be tested and proven reliably safe. Then, the test payload used on the RC paraglider will be refitted for the small-scale rocket. The payload that will be integrated into the first small-scale test rocket has already been designed and is seen in Figure 7. The small scale rocket will undergo multiple tests in order to gather a significant amount of flight data and feedback about the problems and successes of the autopilot system. This stage of testing will be most representative of flight performance and characteristics of the payload that will be used in a large-scale rocket. In about a year, this data will be used to begin

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development of the preliminary design of a large-scale, level 3 or larger rocket. These are the types of rockets are used in international rocket competitions. This is where the true value of the Polaris guided parachute system will be tested.

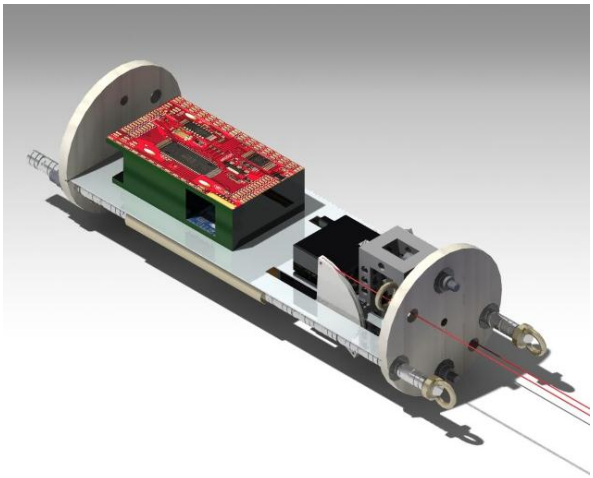

Figure 7: Early-stage design of the guided parachute payload to be used in a 4" test rocket

Because this project has been so recently developed, younger ERPL members have been given the opportunity to participate in early project design and development. Most all other projects being run by ERPL are several years into operations and have moved beyond their respective initial design and planning phases. Polaris members are primarily focused on research and the initial design of the payload and its testing systems. As this project moves into the testing phases, members will be in charge of developing test procedures and documents to ensure that tests are completed thoroughly and efficiently while getting the results that this project needs. While there is a project lead that technically runs the project, responsibilities and leadership roles are spread out evenly among the small team that is developing this process, meaning that every member has meaningful responsibilities that contribute greatly to the success of this project.

Polaris is a partner project, meaning both ERPL and a similar rocketry organization called the Embry-Riddle Future Space Explorers and Developers Society (ERFSEDS) are working on this project together. The end goal of this project is to have the autonomous parachute be a payload that can be easily integrated into any rocket in either club. From small-scale rockets to large-scale rockets, the point is to increase reliability and recoverability of rockets used for experiments, competitions, or anything in between. Over the years, many rockets have been lost or damaged thanks to unreliable recovery systems. Polaris aims to be the solution to this problem. With this, future experiments and competitive rockets will always have a reliable and convenient recovery system at hand.

V. Conclusion

ERPL offers students a variety of high-level projects in rocket subsystem development. The main goal of these projects is to encourage students to grow and develop skills in multiple engineering disciplines through designing, building, and testing rocketry subsystems so that they can bring hands-on experience and knowledge to industry when they graduate. Members experience many issues through the process of designing these systems and are in charge of solving these problems in order to achieve success in their projects. Already, ERPL offers projects that cover topics such as reliability, modeling and simulation, systems verification and validation, autonomous systems and model-based systems engineering. In the future, ERPL hopes to support more projects created by students in order to ultimately increase the variety of engineering disciplines that members can be exposed to and learn from.

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